

ECE 447

Lesson 25

Digital Receivers: Timing, Detection, & Eye Diagrams



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **Lab 5.** Was due Lesson 24 (15 Oct).
 - **HW6.** Due Lesson 28 (Tue, 27 Oct) to Gradescope.
 - **HW7.** Assigned Lesson 26 (Wed, 21 Oct); due Lesson 32 (Fri, 6 Nov).

REVIEW

- What is Nyquist's First Criterion?
- True or False. For error-free communication, we need zero ISI across the pulse width, T_b .
- f_x is the excess bandwidth. In excess of what value?

SCRAMBLING

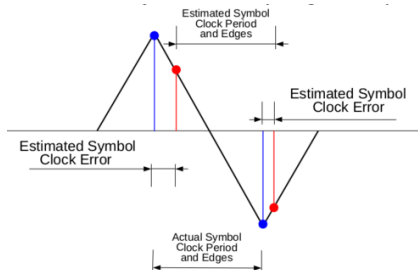
- Section 6.4 - not part of your reading assignments
- Uses shift registers to mix up the 1's and 0's
- Makes the data appear random - helpful for timing extraction and security purposes
- Not the same as encryption

EQUALIZERS

- Section 6.5.1 - also not part of your reading assignments
- Received signals usually attenuated and distorted by the channel
 - how do we compensate for the attenuation?
- Distortion is compensated by an equalizer
- Goal is to mitigate ISI while simultaneously suppressing the channel noise - but you can't do both at the same time
- Digital signals, however, don't need to worry about fully mitigating ISI
- Equalizer types: zero-forcing, MMSE, adaptive
- Lab 5 uses an adaptive, *blind* equalizer in the flowgraph (see Section 11.10 for more on blind equalization)

TIMING EXTRACTION

- Precise synchronization critical for symbol detection - 3 approaches are:
 1. Derive from a primary or secondary clock (e.g., GPS or a German atomic clock broadcasting as a master timing source)
 2. Send as separate signal (pilot)
 3. Self-clocking signal (ideal)
- Lab 5 - uses Symbol Sync block for timing synchronization, signal decimation, and filtering



Symbol Sync

Timing Error Detector: Mueller and Müller

Samples per Symbol: 8

Expected TED Gain: 1

Loop Bandwidth: 45m

Damping Factor: 1

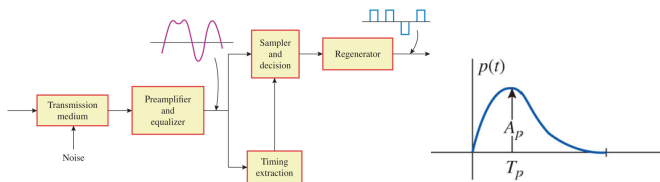
Maximum Deviation: 1.5

Output Samples/Symbol: 1

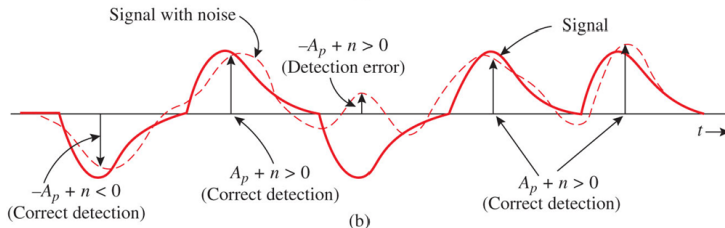
Interpolating Resampler: MMSE, 8 tap FIR

DETECTION

- Occurs after equalization and timing extraction
- Received signal = equalized pulse train + AWGN



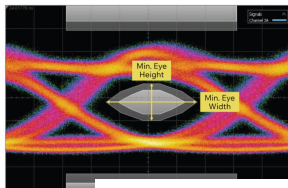
(a)



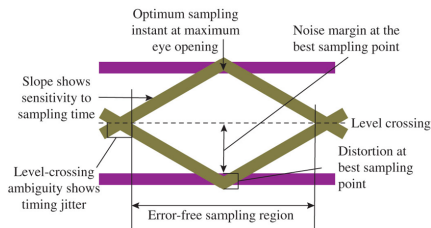
(b)

EYE DIAGRAMS

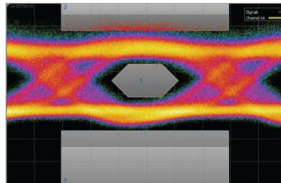
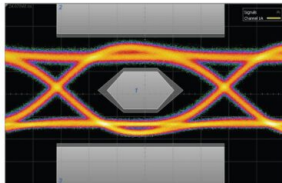
- Practical engineering tool used to diagnose level of noise, tolerance to jitter/timing issues, level of jitter, noise margin, rise/fall times
- Eye diagrams in practice



Stable Eye Diagram

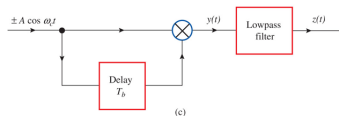
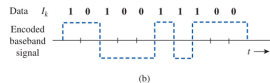
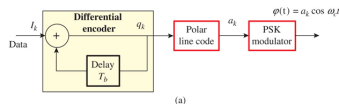


Unstable Eye Diagram



DIFFERENTIAL ENCODING

- Lab 5 uses DQPSK, or Differential Quadrature Phase Shift Keying
- Receiver only looks at relative difference between successive pulses vs. absolute phase
- Allows for noncoherent demodulation



Time k	0	1	2	3	4	5	6	7	8	9	10
I_k		1	0	1	0	0	1	1	1	0	0
q_k	0	1	1	0	0	0	1	0	1	1	1
Line code a_k	-1	1	1	-1	-1	-1	1	-1	1	1	1
θ_k	π	0	0	π	π	π	0	π	0	0	0
$\theta_k - \theta_{k-1}$	π	0	0	π	0	0	π	π	π	0	0
Detected bits	1	0	1	0	0	1	1	1	0	0	